

Trevor Smith
Professor Ashton
ISTA 421
04.28.26

Part C: Final Project Outline

Research Question: Do machine learning models used to predict NYC taxi fares reveal systematic biases or inconsistencies across geographic locations and trip characteristics, and which predictors contribute most to these disparities?

1. Problem Formulation and Significance

1.1 Problem Statement

- NYC taxi fares follow standardized rules but vary in ways that are not always transparent to riders
- ML fare prediction models may perform unevenly across geographic zones and trip types
- ML task: Supervised regression (predicting fare_amount)

1.2 Significance

- Identifying pricing inconsistencies supports fairness and transparency in urban transportation
- Extends beyond accuracy to interpretability and bias analysis a real-world ML concern
- Relevant to data science applications in civic tech, transportation equity, and policy

1.3 Dataset Description

- Source: NYC Taxi & Limousine Commission (TLC) — yellow_tripdata_2026-01.parquet
- 4,181,444 observations; 20 features
- 12 quantitative features: passenger_count, trip_distance, fare_amount, tip_amount, tolls_amount, total_amount, extra, mta_tax, improvement_surcharge, congestion_surcharge, Airport_fee, cbd_congestion_fee
- 6 qualitative features: VendorID, RatecodeID, store_and_fwd_flag, PULocationID, DOLocationID, payment_type
- Target variable: fare_amount

2. Exploratory Data Analysis

2.1 Data Exploration

- Summary statistics (mean, std, min/max) for fare_amount, trip_distance, tip_amount, total_amount, passenger_count
- Correlation heatmap of all quantitative features vs. fare_amount
- Discuss which features are most correlated with the target

2.2 Handling Missing / Imbalanced Data

- Drop or flag rows where passenger_count = 0 (likely data entry errors)
- Remove negative fare_amount values and extreme outliers (e.g., fares > \$500)
- Check for and handle NaN values across all features
- Normalize/standardize continuous features prior to linear regression

2.3 Data Visualizations

- Histogram of fare_amount distribution
- Histogram of trip_distance distribution
- Scatterplot: trip_distance vs. fare_amount
- Boxplot of fare_amount by PULocationID (top 10 busiest zones) to show geographic variation

3. Model Selection, Application, and Evaluation

3.1 Model Justification

- Linear Regression (baseline): Interprets linear relationships between trip features and fare; transparent and interpretable; written from scratch using normal equations or gradient descent
- Decision Tree Regressor (comparison): Captures non-linear relationships; produces feature importance scores natively; well-suited to detecting location-based patterns
- Both models chosen to test hypothesis: global models may underperform for specific geographic subsets

3.2 Method Application

- Features used: trip_distance, passenger_count, PULocationID, DOLocationID, congestion_surcharge, cbd_congestion_fee, tolls_amount, Airport_fee
- Train/test split: 80/20
- Linear Regression: implemented from scratch (no sklearn fit); verified against sklearn
- Decision Tree: hyperparameter tuning via cross-validation max_depth, min_samples_leaf

- All generative AI usage will be labeled in code comments and shared with Professor Ashton

3.3 Model Evaluation

- Metrics: RMSE, MAE, R^2
- Compare Linear Regression vs. Decision Tree across all three metrics
- Feature importance analysis from Decision Tree
- Residual analysis: plot residuals by PULocationID and DOLocationID to detect geographic bias

4. Results, Conclusions, and Real-World Implications

4.1 Results Presentation

- Table comparing RMSE, MAE, R^2 for both models
- Bar chart of feature importance scores from Decision Tree
- Residual plots by pickup zone identifying which locations have highest prediction error
- Tie all results back to the research question and hypothesis

4.2 Conclusions

- Interpret whether hypothesis is supported: do global models underperform for specific locations?
- Discuss which features most influence fare prediction and what that implies for pricing fairness
- Limitations: single month of data; no time-of-day features; location IDs not geographically labeled

4.3 Workforce / Graduate School Preparation

- Skills developed: EDA on large real-world datasets, supervised ML implementation from scratch, bias/fairness analysis, Python (pandas, matplotlib, sklearn), technical communication
- Discuss relevance to career or graduate school goals

5. Oral Presentation Structure

- ~1 min — Problem statement and research question
- ~1 min — Significance: why fare bias matters
- ~1.5 min — Dataset overview (source, size, features)
- ~2 min — EDA highlights: 2–3 key visualizations
- ~2 min — Model selection, implementation, and hyperparameter tuning
- ~1.5 min — Results: metrics table, feature importance, residual maps

- ~1 min — Conclusions, limitations, and real-world implications
- Q&A from classmates and Professor Ashton

6. Code Implementation and Documentation

- Jupyter Notebook (taxi_analysis.ipynb) fully runnable end-to-end pipeline
- Sections: data loading, EDA, preprocessing, model implementation, evaluation, visualization
- All figures in the paper generated by the code
- README.md with instructions for running the code
- Dataset submitted separately on D2L (not GitHub)